

Section A: Rajagiri Academic System

Create a database schema for a college academic management system. The system should manage students, departments, faculty, courses, enrollments, results, and events.

Table 1: DEPARTMENT

Store the following details:

- Department ID
- Department Name
- Head of Department
- Office Location
- Contact Number

Constraints:

- Department ID should be the primary key.
- Department Name should be unique and not null.

Column Name	Description
dept_id	Department ID
dept_name	Department Name
hod_name	Head of Department
office_location	Office Location
contact_number	Department Contact Number

Table 2: STUDENT

Store the following details:

- Student ID
- Student Name
- Date of Birth
- Gender
- Mobile Number
- Email ID
- Department ID
- Admission Date
- CGPA

Constraints:

- Student ID should be the primary key.
- Email ID should be unique.
- Gender should allow only M or F.
- CGPA should be between 0 and 10.
- Department ID should refer to the DEPARTMENT table.

Column Name	Description
student_id	Student ID
student_name	Student Name
dob	Date of Birth
gender	Gender
mobile_number	Mobile Number
email_id	Email ID
dept_id	Department ID
admission_date	Admission Date
cgpa	CGPA

Table 3: FACULTY

Store the following details:

- Faculty ID
- Faculty Name
- Qualification
- Designation
- Date of Joining
- Salary
- Email ID
- Department ID

Constraints:

- Faculty ID should be the primary key.
- Email ID should be unique.
- Salary should be greater than zero.
- Department ID should refer to the DEPARTMENT table.

Column Name	Description
faculty_id	Faculty ID
faculty_name	Faculty Name
qualification	Qualification
designation	Designation
joining_date	Date of Joining
salary	Salary
email_id	Email ID
dept_id	Department ID

Table 4: COURSE

Store the following details:

- Course Code
- Course Name
- Credits
- Semester
- Department ID
- Faculty ID

Constraints:

- Course Code should be the primary key.
- Course Name should not be null.
- Credits should be between 1 and 6.
- Semester should be between 1 and 8.
- Department ID should refer to the DEPARTMENT table.
- Faculty ID should refer to the FACULTY table.

Column Name	Description
course_id	Course ID
course_name	Course Name
credits	Number of Credits
semester	Semester
dept_id	Department ID
faculty_id	Faculty ID

Table 5: ENROLLMENT

Store the following details:

- Enrollment ID
- Student ID
- Course Code
- Enrollment Date
- Academic Year

Constraints:

- Enrollment ID should be the primary key.
- Student ID should refer to the STUDENT table.
- Course Code should refer to the COURSE table.
- Enrollment Date should default to the current date.
- A student should not enroll in the same course more than once.

Column Name	Description
enrollment_id	Enrollment ID
student_id	Student ID
course_id	Course ID
enrollment_date	Enrollment Date
academic_year	Academic Year

Table 6: RESULT

Store the following details:

- Result ID
- Enrollment ID
- Internal Marks
- External Marks
- Total Marks
- Grade
- Result Status

Constraints:

- Result ID should be the primary key.
- Enrollment ID should refer to the ENROLLMENT table.
- Internal Marks should be between 0 and 50.
- External Marks should be between 0 and 50.

- Grade should allow values such as A, B, C, D, F.
- Result Status should allow only PASS or FAIL.

Column Name	Description
result_id	Result ID
enrollment_id	Enrollment ID
internal_marks	Internal Marks
external_marks	External Marks
total_marks	Total Marks
grade	Grade
result_status	PASS/FAIL

Table 7: EVENT

Store the following details:

- Event ID
- Event Name
- Event Date
- Venue
- Coordinator Faculty ID
- Maximum Participants
- Registration Fee

Constraints:

- Event ID should be the primary key.
- Coordinator Faculty ID should refer to the FACULTY table.
- Maximum Participants should be greater than zero.
- Registration Fee should not be negative.

Column Name	Description
event_id	Event ID
event_name	Event Name
event_date	Event Date
venue	Venue
faculty_id	Coordinator Faculty ID
max_participants	Maximum Participants
registration_fee	Registration Fee

Table 8: EVENT__REGISTRATION

Store the following details:

- Registration ID
- Event ID
- Student ID
- Registration Date
- Participation Status

Constraints:

- Registration ID should be the primary key.
- Event ID should refer to the EVENT table.
- Student ID should refer to the STUDENT table.
- Registration Date should default to the current date.
- Participation Status should allow values REGISTERED, ATTENDED, ABSENT.

Column Name	Description
registration_id	Registration ID
event_id	Event ID
student_id	Student ID
registration_date	Registration Date
participation_status	REGISTERED/ATTENDED/ABSENT

ALTER TABLE Assignment: Replace HOD Name with Faculty Foreign Key

Question

In the existing department table, the Head of Department is stored as hod_name. This causes data redundancy because faculty details are already stored in the faculty table.

Modify the database design so that each department stores the HOD using faculty_id instead of hod_name.

Perform the following tasks:

1. Add a new column named hod_faculty_id to the DEPARTMENT table.
2. Create a foreign key relationship between hod_faculty_id and faculty_id in the FACULTY table.
3. Remove the existing hod_name column from the DEPARTMENT table.

4. Display the structure of the modified DEPARTMENT table.